Week 50: Staircases and lengths

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Mathematical overview

Length invariant. Every finite horizontal/vertical path from S=(0,0) to F=(160,120), using only right and up moves, has horizontal total 160 mm and vertical total 120 mm. Its length is therefore **280 mm**, regardless of the number or sizes of its pieces. Zero-length pieces do not create turns; adjacent collinear pieces should be merged when counting.

The straight diagonal has length **200 mm**, since $160^2 + 120^2 = 200^2$. Staircases can be made arbitrarily close to it while retaining length 280 mm. Here "close" means the largest **vertical** gap $|y - 3x/4|$ over the whole path, not the length along the route or a perpendicular distance. Closeness of curves does not imply closeness of lengths. No finite staircase becomes the diagonal.

If leftward travel is allowed but downward travel is not, let L be the total distance traveled left. Rightward travel must be $160 + L$, so total length is **280+2L mm**. This counts repeated traversal as traveled length; it is not the area or length of the inked set.

Discovery by band. K-1 compares string lengths and makes close or backtracking routes. Grades 2-3 explains totals and minimum turns. Grades 4-5 separates approximation of location from approximation of length. Measurement suggests equality; regrouping all horizontal and vertical pieces proves it.

Prepare and launch

Per pair: student pages, pencils, a ruler, two nonstretch strings at least 40 cm long, and removable tape. Print at actual size. Main rectangles are **160 mm wide by 120 mm high**; verify the printed 100 mm calibration before using string or a ruler. The diagram in the key is deliberately smaller, not a measuring mat.

Readiness. K-1 needs right/up directions and side-by-side comparisons, with adult reading. Older entry requires addition of lengths; explaining the eight-turn lower bound needs control of a vertical-gap constraint. Children need not derive a square-root formula. Keep the three adult-led tables and offer the minimum-turn problem as exploration before proof.

Launch, 4 minutes. Let children bend string into a path, then follow the printed 40-by-30 mm example. Match A and B to the horizontal strip and C and D to the vertical strip. Show that string must follow every corner rather than cut across it. Ask, "Can you make a very different route between the same two dots?"

Flexible hour. 0–5 launch; 5–22 build and compare; 22–25 walk an imagined right/up route; 25–43 narrow-strip or turn challenge; 43–55 compare explanations or allow left moves; 55–60 reset. Save a successful route before changing its rules.

Problems and routes

Problem	K–1	2–3	4–5	route
Two different staircases	P1	P1	P1	first
A one-step and a two-step staircase	P2	P2	P2	first
A staircase closer to the diagonal	P3	P3	P3	first
Fewest turns inside the strip	P4	P4	P4	return
Paths that may go left	P5	P5		return
A leftward piece and less string	P6			return
A narrower strip			P5	return
The exact length of a path that goes left once			P6	return
Could more, smaller pieces reach 200 mm?			P7	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P4, P5, P6; Grades 2–3 P4, P5; Grades 4–5 P4, P5, P6, P7.

Key, one entry per problem

Two different staircases

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route

Idea. Every right-and-up path from S to F has length width + height = 280 mm.

Any two different right/up routes between the printed endpoints work. For example, right 160 then up 120, and up 60, right 80, up 60, right 80. Each is 280 mm. Small differences in measured strings are measurement error, not a counterexample to the segment-total proof.

A one-step and a two-step staircase

K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. Regrouping horizontal and vertical pieces proves the two lengths equal.

Both solid and dashed routes are **280 mm**. Solid: 160+120. Dashed: 60+80+60+80. K-1 should circle “same length.” The two paths need not look equally direct for their horizontal and vertical totals to agree.

A staircase closer to the diagonal

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

Idea. Staircases can get as close to the diagonal as you like, yet every one stays 280 mm long; the diagonal is 200 mm.

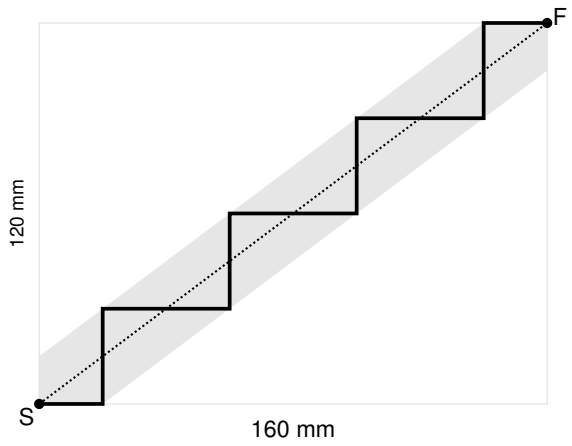
The given four-step staircase reaches a greatest vertical gap of **30 mm**. One improvement is eight repetitions of right 20, up 15, whose greatest gap is **15 mm**; it stays within the gray strip, including its boundary. Both staircases have length **280 mm**, whereas the diagonal is **200 mm**. Along each segment the signed gap is linear, so checking its endpoints controls the whole segment.

Fewest turns inside the strip

K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

Idea. Inside the ±15 mm strip the fewest turns is 8; a horizontal piece spans at most 40 mm (20 mm at either end).

The fewest turns is **8**. A boundary-following witness, in millimeters from S, is (0,0), (20,0), (20,30), (60,30), (60,60), (100,60), (100,90), (140,90), (140,120), (160,120). Its nine nonzero segments make eight direction changes. All vertices have vertical gap in $[-15, 15]$; linearity gives the same for the full path.



Guide diagram, not to scale

Why seven turns cannot work. Inside the strip, a horizontal segment can span at most 40 mm: crossing a vertical-gap range of 30 takes $30/(3/4) = 40$. A horizontal segment beginning at S or ending at F can span at most 20 mm. With at most seven turns, there are at most eight alternating horizontal/vertical segments. If there are four horizontal segments, at least one is an endpoint segment, so horizontal travel is at most $20 + 3 \times 40 = 140$ mm; if there are fewer, it is at most 120 mm. Neither reaches the required 160 mm.

Paths that may go left

K–1 P5 · Grades 2–3 P5 · return route · wording differs by band

Idea. With total leftward travel L (and no downward travel) the length is 280 + 2L mm.

Here are two distinct 360 mm routes, given as right (R), left (L), and up (U) distances in millimeters: **Route A:** R80, U40, L40, U40, R120, U40. **Route B:** R120, U90, L40, U30, R80. Both stay in the rectangle, end at F and travel left a total of 40 mm. Each length is $280 + 2 \times 40 =$ **360 mm**. K-1 needs only two equal longer paths, so these give a complete witness without requiring arithmetic.

Grades 2–3: The last question has the same answer as *A leftward piece and less string* below: no path with a leftward piece is shorter than 280 mm.

A leftward piece and less string

K–1 P6 · return route

Idea. Every millimetre left is recovered by an extra millimetre right.

A positive leftward piece cannot make a path shorter than 280 mm. Every millimeter left must be recovered by an extra millimeter right, adding two millimeters to the baseline. A route with any positive leftward travel is strictly longer than a staircase.

A narrower strip

Grades 4–5 P5 · return route

Idea. Closeness in position does not bring the length closer: an 8 mm strip still forces 280 mm.

For the ± 8 mm strip, start U7.5. Then use eight R20 pieces, with U15 between consecutive horizontal pieces, and finish U7.5. This reaches (160,120) and has maximum vertical gap **7.5 mm**. Its length is still **280 mm**, so it cannot be shorter than a route in the wider strip if both obey the same right/up rule. This is a finite witness; no infinite physical construction is required.

The exact length of a path that goes left once

Grades 4–5 P6 · return route

Idea. Net horizontal displacement stays 160 mm, but horizontal distance travelled does not.

The printed left-once route is Route A in the key to *Paths that may go left*. Horizontal travel is 80+40+120=240 mm and vertical travel is 40+40+40=120 mm, giving **360 mm**. Net horizontal displacement remains 160, but horizontal distance traveled is no longer 160; this is the exact part of the staircase argument that changes.

Could more, smaller pieces reach 200 mm?

Grades 4–5 P7 · return route

Idea. The diagonal is a limit of staircases in position, not in length.

No finite right/up staircase can have length 200 mm. Its total horizontal and vertical distances remain 160 and 120. With n equal right-then-up steps, the largest vertical gap is $120/n$ mm, which can be made smaller than any specified positive tolerance by taking a sufficiently large finite n. The length remains 280 mm. The diagonal is a separate limiting curve with length 200 mm, not a last member of the staircase sequence.

Optional hints and facilitation checks

For length, collect horizontal and vertical pieces separately as the opening diagram does. For a strip violation, compare vertically with the diagonal at the same x-position. For minimum turns, ask how far one horizontal piece could possibly go before leaving the strip; do not begin by supplying the optimal coordinates.

Let younger children improve a legal route and count its turns without requiring the lower-bound proof. If string measurements seem unequal, check for stretch, shortcutting corners or inconsistent endpoints. The exact argument is about ideal centerline lengths; printed stroke thickness is not an extra corridor allowance.

Source and verification

John D. Norton, *Badly Behaved Curves*, sections “Proof that $\sqrt{2} = 2$,” “The Limit Curve” and “Badly Behaved Limits of Curves”: author’s exposition (sites.pitt.edu/~jdnorton/teaching/paradox/chapters/measure_curves/measure_curves.html). The 160-by-120 mm staircase version, exact finite corridor witnesses, eight-turn lower bound and backtracking examples here were independently derived and checked. All printed dimensions were checked digitally; string handling and session timing are unpiloted proposals, not reports of physical tests.